

ARTICLES

Chained LLM-human interactive framework for electrolyte design in four-electron Zn-I₂ batteries

Zihui Chen¹, Jinglang Zhang², Guanhua Zhao², Qilong Wang^{2*}, Chunpeng Yang^{1*} & Yu-Guo Guo^{3,4*}¹Tianjin Key Laboratory of Advanced Carbon and Electrochemical Energy Storage, School of Chemical Engineering and Technology, Tianjin University, Tianjin 300350, China²Tianjin Key Lab of Machine Learning, College of Intelligence and Computing, Tianjin University, Tianjin 300350, China³CAS Key Laboratory of Molecular Nanostructure and Nanotechnology, Beijing National Laboratory for Molecular Sciences (BNLMS), Institute of Chemistry, Chinese Academy of Sciences (CAS), Beijing 100190, China⁴University of Chinese Academy of Sciences (UCAS), Beijing 100049, China*Corresponding authors (email: qlwang@tju.edu.cn; cpyang@tju.edu.cn; ygguo@iccas.ac.cn)

Received 9 April 2025; Accepted 22 May 2025; Published online 31 July 2025

Abstract Generative artificial intelligence (AI), particularly large language models (LLMs) such as ChatGPT and DeepSeek, is gaining rapid development and providing powerful tools for chemical and material discovery, even for researchers without expertise in programming. However, the complexity of chemical mechanisms and sparse data characteristics impose difficulties in harnessing LLMs in specific materials experimental research. In this study, we propose a chained LLM-human interaction study to utilize GPT-4 in professional chemistry/material research. Specifically, we build an interactive workflow to break down the task into several steps to propose and scrutinize electrolyte materials for four-electron zinc-iodine (Zn-I₂) batteries, an emerging branch of secondary batteries for energy storage. In the interactive study, GPT-4 retrieves key literature information for human comprehension, recommends promising electrolyte materials for experimental evaluation, and rates the potential materials with humans for battery application. Through the chained LLM-Human interaction, we not only summarize the mechanism and find applicable materials for multi-electron zinc-iodine batteries, enhancing the performance of four-electron Zn-I₂ batteries (more than 1000 cycles with high Coulombic efficiency) but also establish a feasible approach to integrating LLMs in professional experimental research, which could further inspire and accelerate chemistry and materials science.

Keywords large language models, GPT-4, chain-of-thought, zinc-iodine batteries, electrolyte materials

1 Introduction

In recent years, artificial intelligence (AI) has been powering scientific discovery across multiple disciplines [1,2]. Especially, large language models (LLMs), such as GPT-4, have emerged with great potential to revolutionize the way of research in chemistry and materials science [3–10]. For instance, GPTs have been applied and found available in the design and modification of metal-organic frameworks (MOFs), in which they can assist researchers in rapidly mining text, extracting insights, and even predicting synthesis processes [11–14]. Moreover, LLMs run with general language, without requiring in-depth knowledge of computer science or complex theoretical models. This unique feature enables experimental researchers in the fields of chemistry and materials to effectively utilize these AI-powered tools. Therefore, LLMs offer great opportunities for reshaping chemical and material research.

Despite the promise, utilizing LLMs in in-depth experimental research still remains challenging, stemming from both the features of LLMs and the intricacy of chemistry/material experimental research [15]. For complex material design tasks in specific experiments, most LLMs often produce hallucinatory results if simply using Q&A modes (questions and answers), especially for multi-step tasks involving external knowledge, data collection, and result analysis without pretraining and proper prompting. To mitigate the hallucination of LLMs, advanced techniques such as retrieval-augmented generation (RAG), have been proposed and

implemented in LLM-assisted chemical and material research [16,17]. However, RAG relies heavily on structured large-scale databases. Moreover, chain-of-thought (CoT) prompting, which involves a series of intermediate reasoning steps, has shown particular promise in resolving complex reasoning problems like arithmetic reasoning, common sense reasoning, and symbolic reasoning [15,18]. In the fields of chemistry and materials, the inherent complexity and professionalism of experimental data often result in discrete, non-continuous outputs rather than idealized, vectorized datasets. This results in limited availability of high-quality data, which is crucial for model performance, posing significant challenges for LLM-assisted experimental research [19,20]. Thus, while LLMs have been frequently utilized for text-mining and literature analysis for chemistry and materials discovery, directly applying LLMs to assist in specific experimental research remains a grand challenge [21].

In this study, we endeavor to harness LLMs (GPT-4) in solving problems in specific and professional experimental research through a CoT-like chained LLM-human interaction. Specifically, we employ ChatGPT (powered by GPT-4) to assist in investigating electrolyte components in zinc-iodine (Zn-I₂) batteries. Zn-I₂ batteries are recently attracting increasing interest in energy storage due to their high energy-storage efficiency and cost-effectiveness [22,23]. Compared with the traditional Zn-I₂ batteries featuring two-electron conversion (I⁰/I[−]) with relatively low capacity and discharge plateau (1.2 V), multi-electron Zn-I₂ batteries, especially four-

electron Zn-I₂ batteries ($I^+/I^0/I^-$) can achieve high output voltage (1.68 V) and high capacity due to activation of high-valence iodine reactions [24,25]. In the four-electron Zn-I₂ battery system, electrolyte components and additives play a crucial role in stabilizing I^+ through coordination and suppressing its hydrolysis with H₂O during cycling. This not only improves the utilization of active species and enhances specific capacity but also reduces the formation of hydrolysis products such as HOI and IO₃⁻, which in turn helps mitigate H^+ generation and electrode corrosion [26,27]. However, currently reported electrolytes for four-electron Zn-I₂ batteries often use high concentrations, suffering from trade-offs such as high viscosity, reduced conductivity, chemical instability, or complex synthesis. Additionally, there is insufficient research on electrolyte materials and mechanisms for four-electron Zn-I₂ batteries. Thus, exploring more potentially applicable materials for stabilizing I^+ is urgently needed for developing four-electron Zn-I₂ batteries. To this end, we use AI-human collaboration research to utilize the broad knowledge of GPT-4 to identify new applicable materials for four-electron conversion iodine cathodes while overcoming LLMs' shortage in task-specific battery research. This chained interactive study has proven to enhance the efficiency and accuracy of discovering new materials for Zn-I₂ battery applications and offers a robust way to integrate LLMs into professional experimental research.

2 Results and discussion

2.1 GPT-human interactive research workflow

In the workflow of a chained LLM-human interactive study (Figure 1), human researchers select the literature of four-electron Zn-I₂ batteries and harness GPT-4's text-mining and reasoning capability to explore unknown experimental results, utilizing prompt engineering to minimize hallucination. Specifically, GPT-4 performs self-generated prompts to summarize given literature and suggest related electrolyte materials for four-electron Zn-I₂ batteries as a battery research assistant. For literature text mining and information output, we begin by generating a prompt with GPT-4. Based on the tasks and the interactive Q&A, GPT-4 generates an accurate prompt, ensuring that GPT-4 can understand the tasks and focus on key information in the literature in the later process.

We provided the latest relevant research papers on four-electron Zn-I₂ batteries for GPT-4 to extract key information efficiently and accurately. By executing the text-mining prompt on the preselected literature, GPT-4 summarizes the key information for electrolyte components in four-electron Zn-I₂ batteries in a table for human comprehension. This workflow enables GPT-4 to retrieve high-quality external knowledge.

Upon establishing a reliable knowledge base, we analyze the mechanisms and key rules of these electrolyte materials influencing four-electron Zn-I₂ batteries. Based on the mechanism understanding, we take advantage of GPT-4's database to generate suggestions for new materials that fit the rules and could enhance the performance of batteries. The suggested electrolyte materials are also scored by GPT-4 based on the criteria we gave. It should be noted that the prompt should be objective in its formulation without any redundant guidance. The proposed candidate with the highest score is further verified by experimental characterizations, followed by testing and optimizing its electrochemical performance in four-electron Zn-I₂ batteries. Finally, we assess the feasibility of these suggested materials for four-electron Zn-I₂ batteries by comparing their scoring with human evaluations based on experimental validation. By integrating human expertise and experimental data with GPT-4's knowledge expansion capability and creativity, we develop a robust interactive research approach with LLMs to uncover new materials for Zn-I₂ batteries.

2.2 Literature summarizing

Prompt engineering has been frequently used in LLMs (including GPTs) as a powerful toolkit for literature text mining [5,28,29]. To establish knowledge based on previous publications on four-electron Zn-I₂ batteries, we also began by generating prompts, not by ourselves but by guiding GPT-4, thereby maximizing the capability of GPT-4. We carefully designed questions and requests that guide GPT-4 to generate efficient prompts, ensuring the extraction of key information from relevant sections in the literature (Figure 2a). This approach breaks down the final objectives into several explicit sections based on our requests to ensure that the model understands the task and works within a defined range, such as to focus on, the location of key information, and the output format. With the generated prompts (detailed in Supporting Information online),

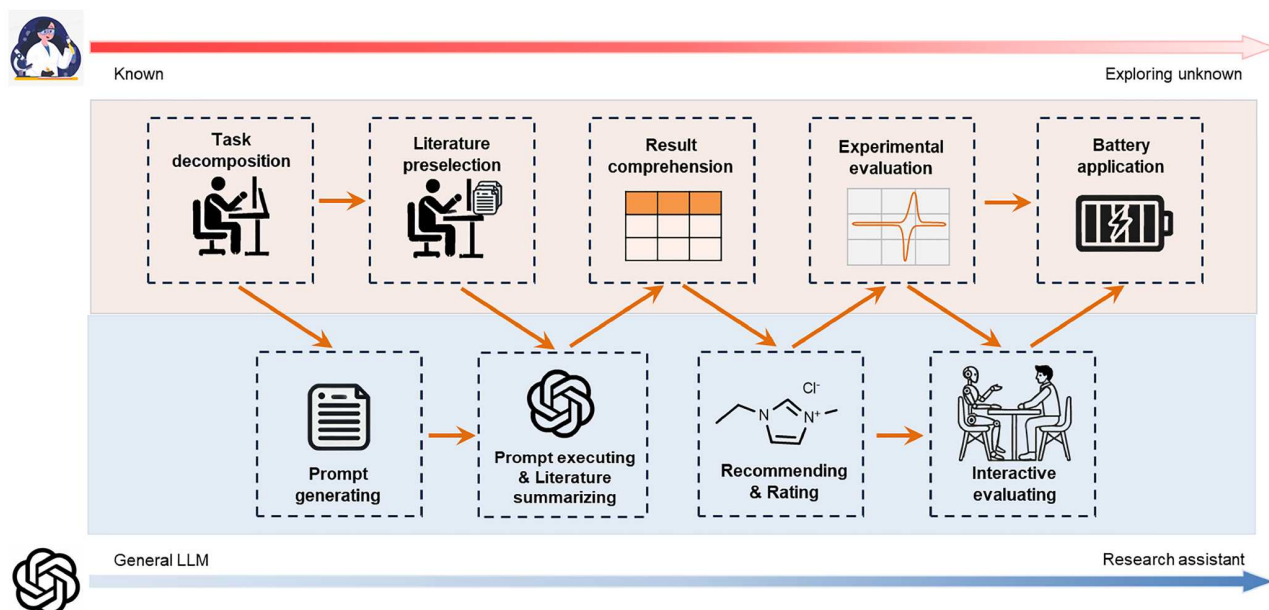


Figure 1 (Color online) The overall workflow of chained LLM-human interactive study for battery research.

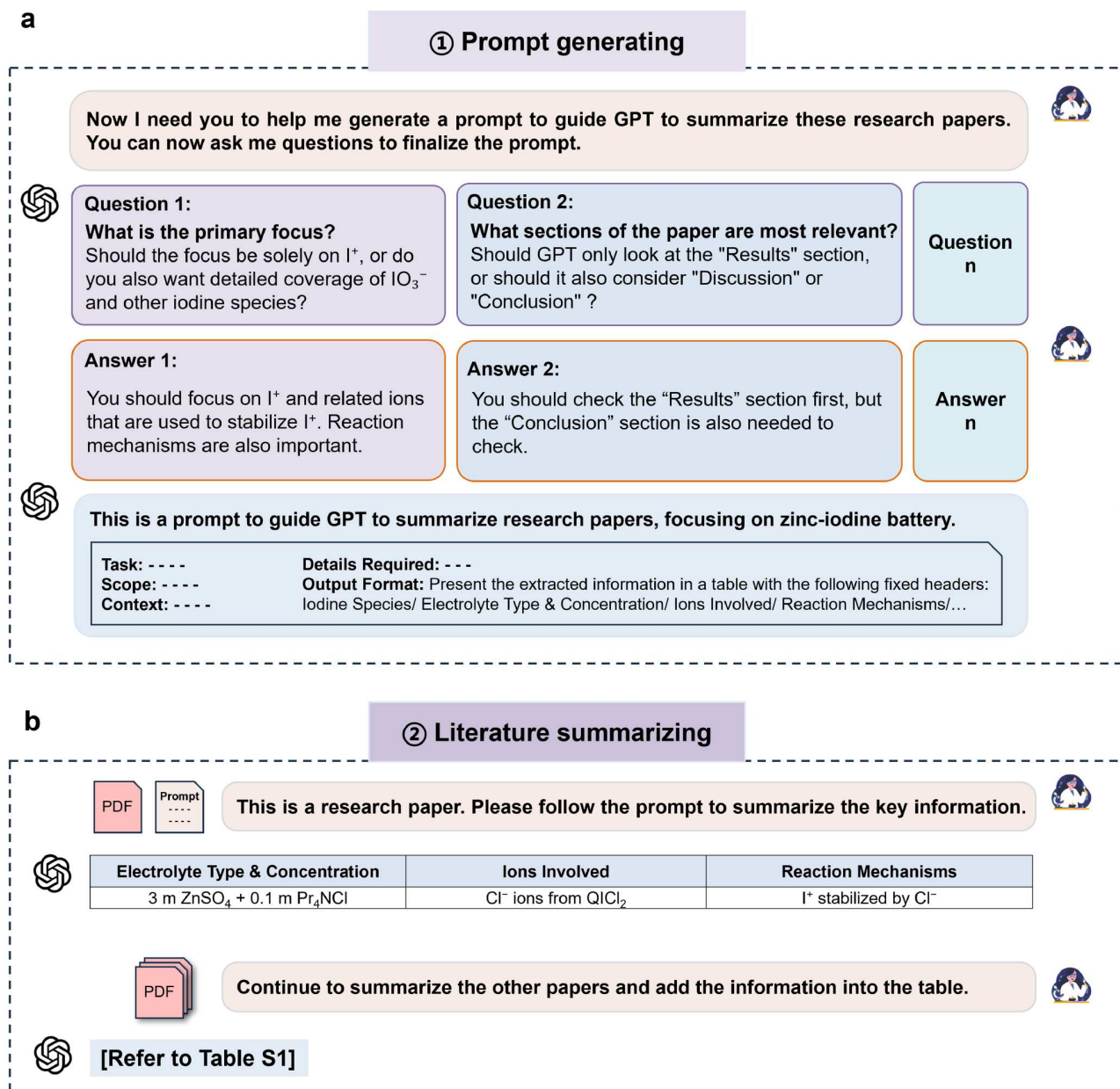


Figure 2 (Color online) Leveraging GPT-4 for literature summary in Zn- I_2 batteries. (a) Self-generated prompt based on question-answering mode. (b) Leveraging GPT-4 to summarize and tabulate the key points for the literature on four-electron Zn- I_2 batteries following the prompt.

GPT-4 rapidly processes about 10 relevant scientific papers on four-electron Zn- I_2 batteries we preselected and then summarizes the findings of the electrolyte materials into a concise, structured table (Figure 2b). Per our prompt, these outputs include critical parameters: electrolyte component, ion types, and reaction mechanisms for four-electron Zn- I_2 batteries (Table 1 and Table S1, Supporting Information online). The comparative table summarized by GPT-4 shows how ions added in the electrolyte, such as Cl^- , Br^- , or tetraethylammonium (TEA^+) reported in the literature, influence I^+ stability and reaction pathways.

This condensed format of literature summary allows us to quickly comprehend the literature findings and grasp the key functioning factors and mechanisms. Analyzing Table 1 [24,25,30–37], we find that the reported electrolyte materials mainly function through two mechanisms: reducing water activity and stabilizing I^+ through coordination (Table S2). For example, PEG200 disrupts hydrogen bonding in water, lowering water activity and protecting I^+ .

Quaternary ammonium salts (Pr_4NCl , TEA^+) create a water-deficient, chloride-rich interface, further stabilizing I^+ . In addition, chloride ions (Cl^-) and bromide ions form interhalogen complexes with I^+ to ICl and $[IBr_2]^-$, preventing capacity decay during redox reactions. Niacinamide (NA) ligands and trimethylamine hydrochloride (TAH) also work similarly, forming stable $I(NA)_2^+$ and bidentate coordination complexes to support reversible redox reactions and prevent hydrolysis. The key mechanism is important to guide GPT-4 for searching and suggesting potential electrolyte components or additive materials.

2.3 Recommending, rating, and experimental evaluating

With comprehension of the key mechanisms for electrolyte components of four-electron Zn- I_2 batteries (i.e., reducing water activity in the electrolyte and combining with the reactive intermediate I^+ species), we used the broad knowledge of GPT-4

Table 1 Key points in the published literature of Zn-I₂ batteries summarized by GPT-4

Electrolyte components	Ions involved	Reaction mechanisms
3 M ZnSO ₄ + 0.1 M Pr ₄ NCl [32]	Cl [−] from QICl ₂	I ⁺ stabilized by Cl [−]
ZnCl ₂ + KCl [33]	Cl [−] ions	I ⁺ stabilized by Cl [−]
ZnSO ₄ + NaCl [34]	Cl [−] , ligand molecules	I ⁺ facilitated by TAH coordination
Zn(ClO ₄) ₂ ·6H ₂ O, dimethyl sulfone (DMS), NA [25]	NA, ClO ₄ [−] ions	I ⁺ stabilized by NA
CuSO ₄ /CuCl ₂ electrolyte [35]	Cu ²⁺ , Cl [−] ions	I ⁺ stabilized by Cl [−]
2 M ZnCl ₂ + PEG200 [24]	Cl [−] ions, PEG	I ⁺ stabilized by Cl [−] , PEG reduces the water activity
1 M ZnSO ₄ + 0.2 M ZnBr ₂ [30]	Br [−] ions	I ⁺ stabilized by Br [−]
2 M ZnCl ₂ + 1 M ZnI ₂ + 0.2 M CoCl ₂ [36]	Co ²⁺ , Cl [−] ions	I ⁺ stabilized by Cl [−]
15 M ZnCl ₂ + TEA ⁺ [37]	Cl [−] ions, TEA ⁺ ions	TEA ⁺ creates a water-poor and Cl [−] -rich interface
10 M ZnCl ₂ , 0 M LiCl, 8 M acetonitrile [31]	Cl [−] , Fe-N ₄ moieties	Fe _{SA} -NC catalyst enhancing the stability of I ⁺

to search for new potential materials that can be applied for four-electron Zn-I₂ batteries by referring to existing materials properties. We guided GPT-4 to provide reliable suggestions step by step (Figure 3a). Specifically, we first asked GPT-4 to categorize the materials in previous reports into two types: reducing water activity and combining with I⁺. Subsequently, we employed these materials as exemplary answers that offer a reference.

We prompted GPT-4 to recommend materials that meet the two critical criteria, explain corresponding mechanisms, and analyze potential side effects. Finally, GPT-4 suggested using non-aqueous solvents, *e.g.*, dimethyl sulfoxide (DMSO) or propylene carbonate (PC), and ionic liquids, *e.g.*, 1-ethyl-3-methylimidazolium chloride ([EMIM]Cl), to lower the water activity in the electrolyte. Additionally, GPT-4 also proposed candidate materials for stabilizing I⁺ through chemical bonding or coordination, such as thiocyanate ions (SCN[−]) and aromatic compounds.

To select the reliable material for experimental verification, we instruct GPT-4 to list the potential electrolyte materials in a table, each with scores rated by several evaluation criteria (Figure 3b). These detailed criteria were made based on the key mechanisms (reducing water activity and combining with I⁺) and additionally, fewer side effects in the batteries (Table S3). For instance, GPT-4 gave [EMIM]Cl a score of 4 (maximum 5) for its ability to reduce water activity. Additionally, GPT-4 rated [EMIM]Cl 4 (maximum 5) for its capability of combining with I⁺ due to Cl[−], which is consistent with the previously reported mechanism [30,31]. However, [EMIM]Cl may have some side effects, such as reducing the ionic conductivity, leading to a score of 2 in terms of fewer side effects (maximum 5). As a result, [EMIM]Cl accumulates a total score of 10 in the GPT-4 scoring system with a full score of 15. We note that although there are certainly other criteria for selecting electrolyte materials, we manually selected the three criteria based on the summarized mechanisms and our expertise, without applying too many restrictions to maximize the creativity of GPT-4.

To experimentally evaluate the potential electrolyte materials proposed by GPT-4, we chose [EMIM]Cl that GPT-4 rated with a high score and tested its corresponding properties by Raman, galvanostatic charge-discharge test, cyclic voltammetry (CV), and electrochemical impedance spectroscopy (EIS). Firstly, we prepared 4 M [EMIM]Cl and 1 M ZnSO₄ as the electrolyte and 4 M NaCl and 1 M ZnSO₄ as a control sample [38]. The Raman spectra (Figure 3c) show a broad band in 3100–3800 cm^{−1}, corresponding to the O–H stretching in the [EMIM]Cl-based electrolyte and NaCl-based electrolyte. The peak at 3296 cm^{−1} (blue part) corresponds to free water with a strong hydrogen bonding network, while the peaks at 3455 and 3565 cm^{−1} (green part) are associated with weakly hydrogen-bonded water [39,40]. Compared with the NaCl-based electrolyte, the intensity of the strong hydrogen bonding in the

[EMIM]Cl-based electrolyte is significantly reduced. Thus, the [EMIM]Cl-based electrolyte contains less free water and less water activity, possibly because the larger ion size of EMIM⁺ can form an effective steric obstruction [37]. Then we used the galvanostatic charge-discharge test (Figure 3d) and CV (Figure S1, Supporting Information online) to evaluate the reversibility of redox reactions of the iodine cathode. The discharge curve of [EMIM]Cl-based Zn-I₂ battery shows a much longer high-voltage plateau, corresponding to I⁺/I⁰ conversion. This is consistent with CV results showing the higher current density of I⁺/I⁰ reduction, smaller over-potential and more symmetric redox peaks in the [EMIM]Cl-based battery compared with the NaCl-based. While the EIS data of Zn-I₂ batteries (Figure S2) show that [EMIM]Cl-based electrolyte increases the electrolyte resistance as predicted by GPT-4 (side effects), the resistance within a reasonable range allows for its application in batteries. The results of ionic conductivity also indicate that [EMIM]Cl-based electrolyte (42.1 mS cm^{−1}) has the potential for battery application (Figure 3e). Additionally, the result of linear sweep voltammetry (LSV) (Figure S3) shows that [EMIM]Cl-based electrolyte has an electrochemical window as wide as 3.5 V, ensuring it can maintain a relatively stable state within the charge-discharge voltage range of 0.5–1.8 V.

In parallel with GPT-4's rating, we rated the performance of [EMIM]Cl based on experimental evaluation criteria: the capacity of the I⁺/I⁰ conversion plateau and fewer side effects (Figure 3f). The length of the high-voltage plateau (I⁺/I⁰ conversion) is used to assess the comprehensive capability of reducing water activity and combining it with I⁺. Based on the experimental results, we gave [EMIM]Cl 8 points in this item (maximum 10) in light of its capability to stabilize I⁺ for high-valence iodine cathodes. In terms of side effects, we found it is stable during the charge-discharge process and then we gave it a score of 4 (maximum 5 for the fewest side effects). Thus, experimental evaluation, albeit on a rough basis, indicates that the [EMIM]Cl recommended by GPT-4 and its underlying logic are generally trustworthy.

2.4 Battery application

We first analyze the functioning mechanism of [EMIM]Cl in four-electron Zn-I₂ batteries. In aqueous environments, I⁺ is generally unstable because water reacts with I⁺, forming other iodine species such as IO₃[−], IO[−], and I[−] [41]. This reaction significantly reduces the stability of I⁺ during high-voltage conditions in aqueous electrolytes. The Cl[−] in [EMIM]Cl can combine with I⁺ to form ICl and stabilize I⁺ [33]. In the meantime, EMIM⁺ ion can create an environment with a less strong hydrogen bonding network as indicated by the Raman spectra, leading to fewer side effects between H₂O and ICl (Figure 4a). In this way, more I⁺ takes part in

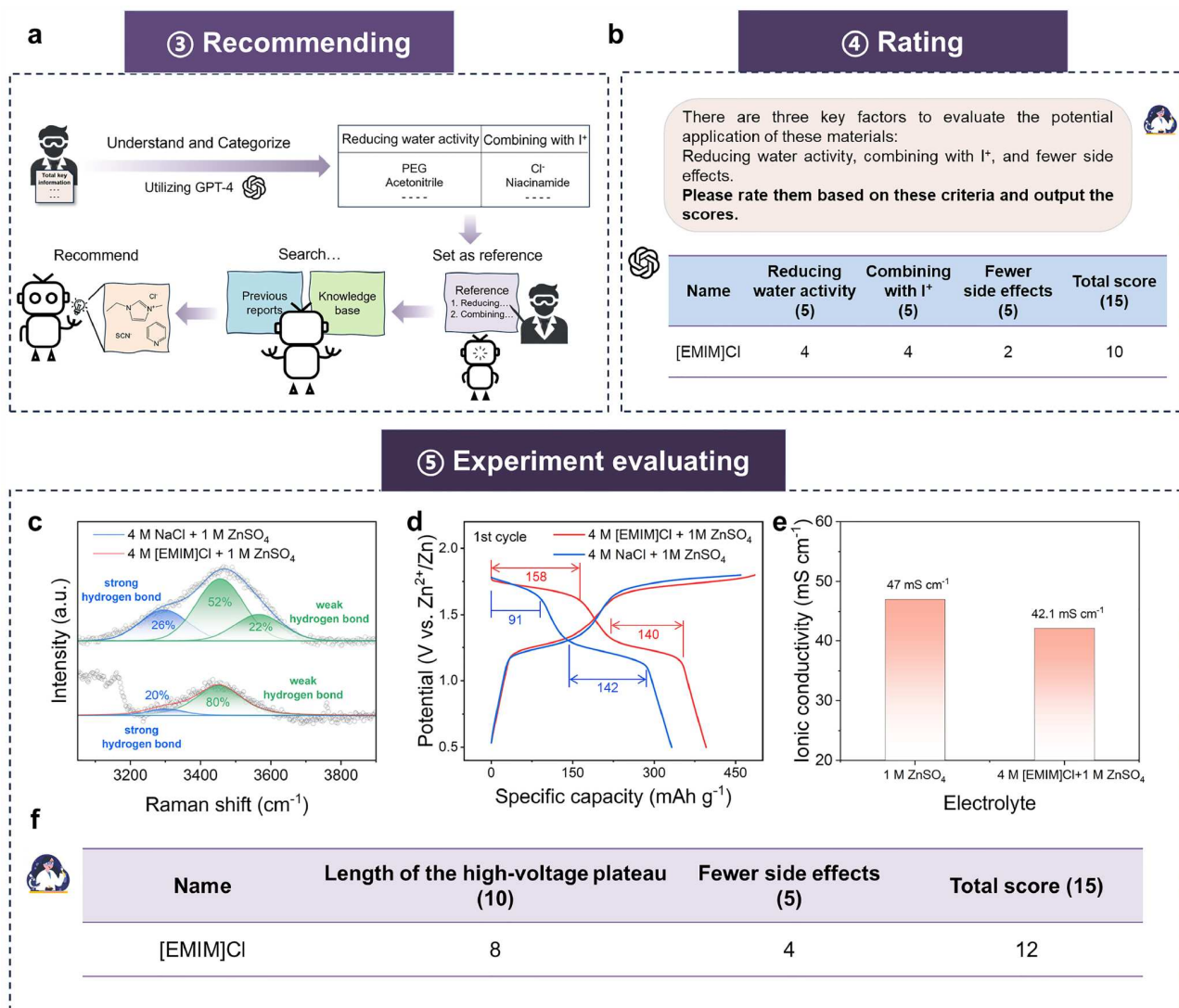


Figure 3 (Color online) Electrolyte material recommending and rating by GPT-4 and experimental evaluation. (a) Leveraging GPT-4 for electrolyte material suggestions for four-electron Zn- I_2 batteries. (b) GPT-4 rating for the suggestions based on three criteria by humans: reducing water activity, combining with I^+ , and fewer side effects. (c) Raman spectra of 1 M $ZnSO_4$ electrolyte with 4 M [EMIM]Cl or 4 M NaCl. (d) Corresponding charge-discharge curves of Zn- I_2 batteries with different electrolytes. (e) Ionic conductivity of 1 M $ZnSO_4$ and 4 M [EMIM]Cl + 1 M $ZnSO_4$ electrolyte. (f) Scores of [EMIM]Cl rated by human researchers based on experimental results of 4 M [EMIM]Cl-based electrolyte.

the electrochemical redox reaction, contributing to a higher utilization rate of I^+ and charge/discharge capacity (particularly in the high-voltage region). For the application of [EMIM]Cl in four-electron Zn- I_2 batteries, we further performed electrolyte component optimization to improve the battery capacity and cycling stability. We added 4 M [EMIM]Cl into 1 M $ZnSO_4$ as the electrolyte, considering the ratio of Cl^- to Zn^{2+} (4:1) can form a coordination structure of $ZnCl_4^{2-}$, which is beneficial to minimize the hydrogen evolution reaction in $ZnSO_4$ electrolyte [38,42]. As shown in cycling performance, the capacity declines to around 150 $mAh\ g^{-1}$ after 150 cycles with 4 M [EMIM]Cl (Figure S4). The possible reason is that too much $EMIM^+$ combined with polyiodide species aggravated the effect of polyiodide shuttling [43,44]. Thus, we further optimized [EMIM]Cl by reducing the concentration of [EMIM]Cl to 1 M and replacing $ZnSO_4$ with $ZnCl_2$ to maintain enough Cl^- .

To experimentally investigate the effect of electrochemical reversibility, we tested CV and calculated Tafel slopes. The CV curve of the Zn- I_2 battery using the [EMIM]Cl-based electrolyte shows much better reversibility between I^0 and I^+ than the one using the NaCl-based electrolyte, indicated by the higher current

density and more symmetric redox peaks (Figure 4b). Notably, the current density of the reduction peak of I^+ in [EMIM]Cl-based electrolyte is twice that of the NaCl-based electrolyte, illustrating that more I^+ can be reduced during the reduction process. The Tafel slopes of the [EMIM]Cl-based electrolyte during iodine oxidation ($169\ mV\ dec^{-1}$) are also found lower than that of the NaCl-based electrolyte (Figure 4c), reflecting the faster reaction kinetics. For the cycling performance of the Zn- I_2 battery, the battery with [EMIM]Cl-based electrolyte shows a higher specific capacity ($315\ mAh\ g^{-1}$) in the first cycle compared to NaCl-based electrolyte ($288\ mAh\ g^{-1}$) (Figure S5). During the cycling, [EMIM]Cl improves the contribution of the high-voltage plateau in charge-discharge curves (Figure 4d and Figure S6), which represents the continuously reversible I^+/I^0 conversion and is critical for long-life, high-energy Zn- I_2 batteries. Specifically, the ratio of the length of the high-voltage plateau to the low-voltage plateau approaches 1:1, which demonstrates that I^0 can convert to I^+ almost completely in [EMIM]Cl-based electrolyte with minimal by-reaction during the charge-discharge process (Figure 4e). In addition, according to the peak location of I^+ and I_2 (341 and 445 nm), UV-vis spectra indicate that more I^0/I^+ conversion occurs in [EMIM]Cl-based electrolyte (Figure S7)

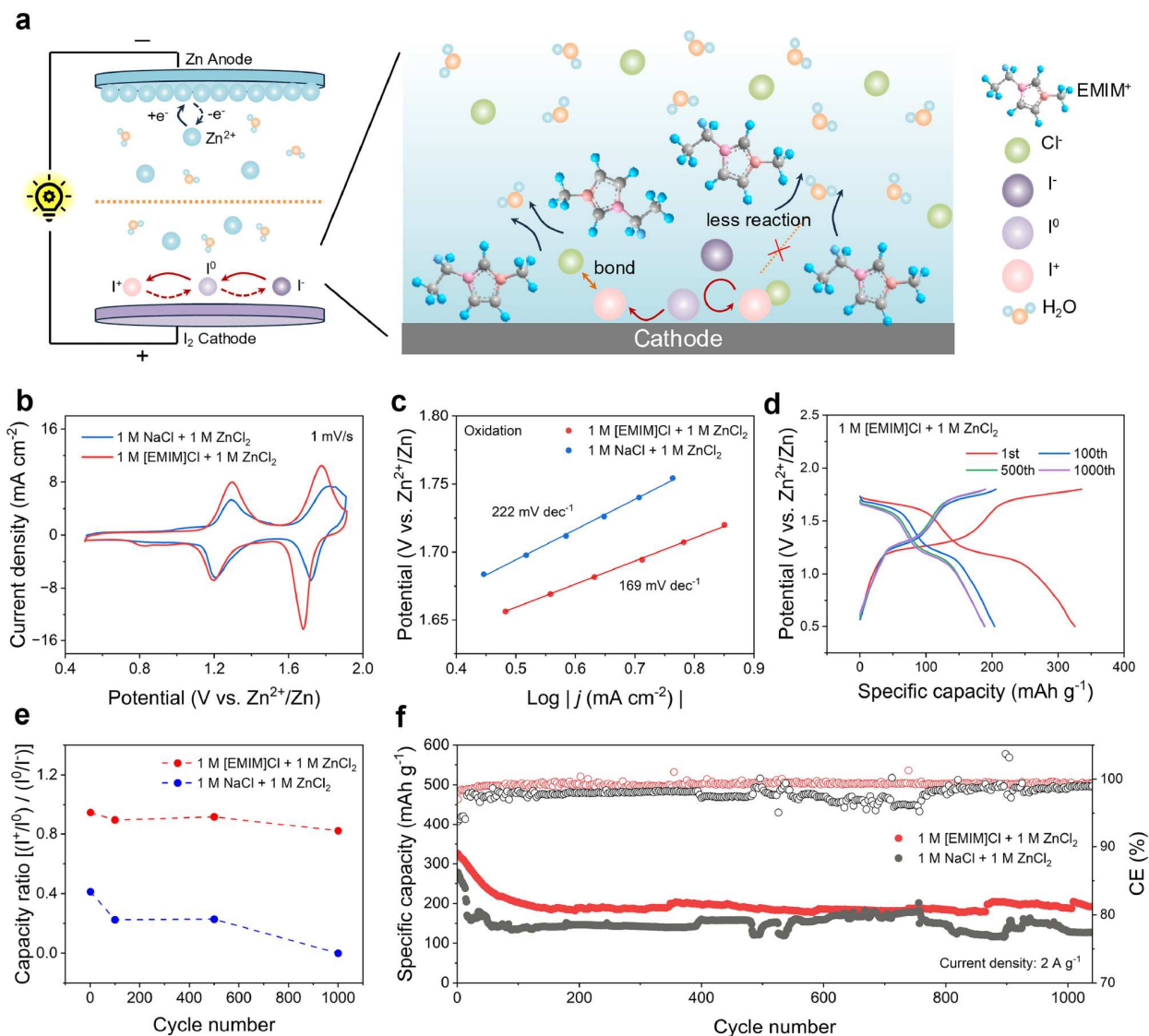


Figure 4 (Color online) The mechanism and performance of Zn-I₂ batteries. (a) Schematics of Zn-I₂ batteries with [EMIM]Cl-based electrolyte. (b) CV curves of Zn-I₂ batteries with 1 M [EMIM]Cl or 1 M NaCl. (c) Tafel plots of Zn-I₂ batteries with [EMIM]Cl during oxidation. (d) Galvanostatic charge-discharge curves of Zn-I₂ batteries with 1 M [EMIM]Cl and 1 M ZnCl₂ during the cycling process. (e) The capacity ratio of high-voltage/low-voltage plateaus [(I⁺/I⁰)/(I⁰/I⁻)] during the cycling process. (f) Cycling stability and CE of Zn-I₂ batteries with [EMIM]Cl.

[40,45]. Specifically, we found that only the I⁺ occurs remarkably when charging to 1.85 V in [EMIM]Cl-based electrolyte, whereas there remains the peak of I₂ and the weak peak of I⁺ when charging to 1.85 V in NaCl-based electrolyte. The possible reason is that I⁺ will break down quickly by side reaction with H₂O in NaCl-based electrolyte, while I⁺ could be maintained longer in [EMIM]Cl-based electrolyte. Therefore, the battery with [EMIM]Cl-based electrolyte shows a higher capacity and more stable cycling performance at a current density of 2 A g⁻¹, achieving more than 1000 cycles with high Coulombic efficiency (CE) (>99.3%), better than the one with NaCl-based electrolyte (Figure 4f). Thus, [EMIM]Cl has proven advantageous in facilitating the I⁺/I⁰ conversion, which enhances the capability of a four-electron conversion Zn-I₂ battery and provides a favorable balance between capacity and voltage (Figure S8, Tables S4 and S5).

2.5 Interactive evaluation for reliability

It is interesting that some electrolyte materials for four-electron Zn-I₂ batteries recommended by GPT-4 have been published during

our research, such as PEG and NA [24,25], which also indicates the reliability of the suggestions. Apart from these published materials, we picked some other materials for further interactive evaluation (Tables S6 and S7) and compared the scores of LLMs (GPT-4 and DeepSeek-R1) obtained through a chained interactive study and ad hoc Q&A (Figure 5a). Specifically, based on the suggestion table listing the potential electrolyte components/additives and corresponding scores given by GPT-4, we selected several materials for experimental evaluation, including NH₄SCN, thiourea, pyridine, and 2-methylimidazole, in addition to [EMIM]Cl. We first verified the rationality of GPT prediction through pre-experiments, including Raman and ionic conductivity and charge-discharge tests (Figures S9–S11). Then we added these materials into the electrolyte of Zn-I₂ batteries and respectively tested their capability of improving the battery performance (enhancing the length of the high-voltage plateau) and reducing side effects. Finally, we rated these electrolyte materials based on the experimental results, in comparison with the scores given by GPT-4 (Figure 5b). It is seen that after the whole chained interactive study, most suggestions by GPT-4 are rather in line with experimental results (Figure 5c). To verify the function of

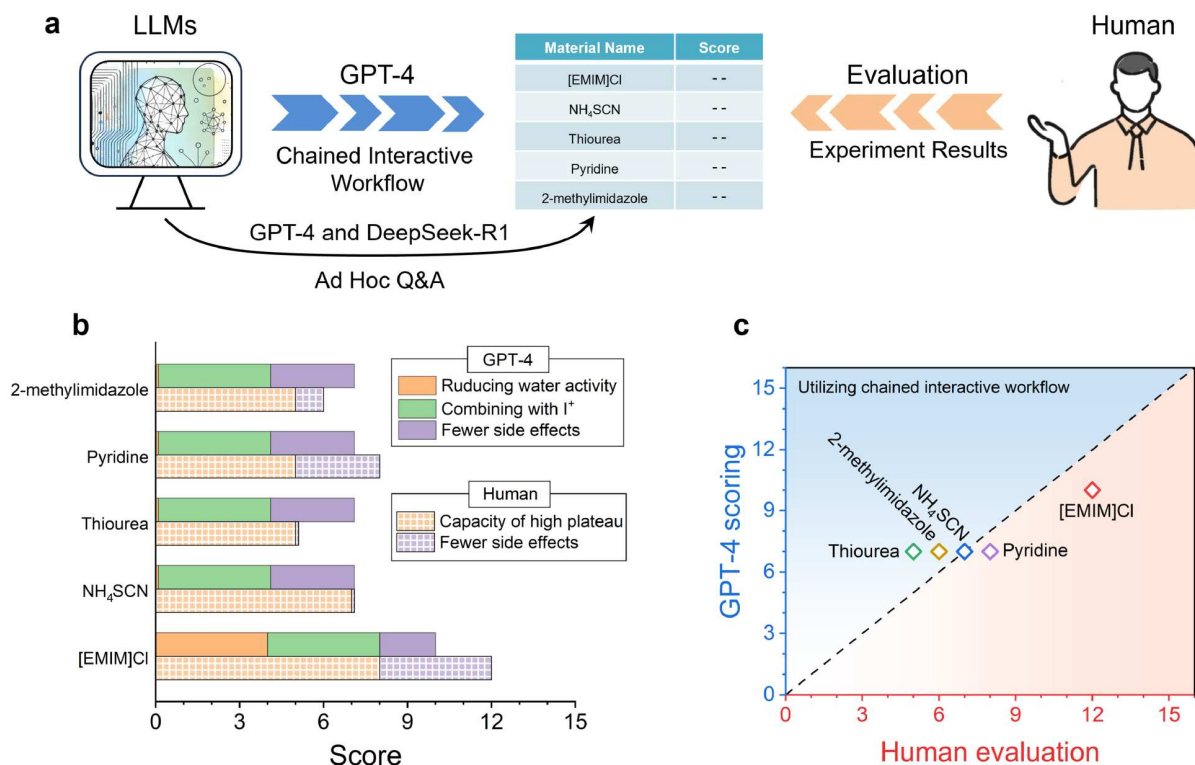


Figure 5 (Color online) Interactive evaluation between humans and LLMs. (a) Scoring strategy through rating by GPT-4 (through the chained interactive study or Ad Hoc Q&A), DeepSeek-R1 (Ad Hoc Q&A), and human (through experimental evaluation). (b) Bar charts comparison of scores and (c) scatter plot comparison of scores for various materials by GPT-4 and human.

chained LLM-human interaction, we also compare the scores given by DeepSeek-R1 and GPT-4 through ad hoc Q&A, without the chained LLM-human interaction (Figure S12a, Tables S8 and S9). Both models produced comparable scoring patterns in the absence of chained interaction, suggesting similar baseline reasoning abilities. However, these outputs significantly deviated from expert evaluations (Figure S12b). This highlights the limitations of static, single-step Q&A and supports the need for human-guided, chained interactive refinement. Therefore, LLMs with chained LLM-human interaction show an acceptable degree of accuracy in recommending potential materials, validating their potential to assist us in exploring materials applications.

3 Conclusions

In summary, by building the chained LLM-human interactive study, we successfully break down the complex task and collaborate with GPT-4 to identify electrolyte materials for four-electron Zn-I₂ batteries. Through the process of literature summarization, material suggestion, and interactive evaluation, we transform GPT-4 into a research assistant to output the key points for human comprehension, identify electrolyte components that can stabilize I⁺, and give corresponding scores. Among these materials, [EMIM]Cl gets a high score and can indeed enhance the conversion between I⁺ and I⁰ and the performance of Zn-I₂ batteries according to our experimental verification. This chained LLM-human interaction not only advances the understanding of Zn-I₂ batteries but also establishes a feasible method for integrating LLMs into experimental research, including other conversion-type cathodes (*e.g.*, sulfur and halogen cathodes) and other materials research. Such a method is accessible to researchers across disciplines with minimal computational expertise requirements for innovation in energy-storage materials and other fields. LLMs participating in interactive scientific research

offer an application paradigm for AI-guided discovery in data-scarce, content-complicated research domains.

Conflict of interest

The authors declare no conflict of interest.

Acknowledgement

This work was supported by the Beijing National Laboratory for Molecular Sciences (BNLMS202404) and the Haihe Lab of Information Technology Application Innovation (22HHXCJC00002). The authors also appreciate the funding support from the Haihe Laboratory of Sustainable Chemical Transformations, the National Industry-Education Integration Platform of Energy Storage, and the Fundamental Research Funds for the Central Universities.

Supporting information

The supporting information is available online at <http://chem.scichina.com> and <http://link.springer.com/journal/11426>. The supporting materials are published as submitted, without typesetting or editing. The responsibility for scientific accuracy and content remains entirely with the authors.

References

- Boiko DA, MacKnight R, Kline B, Gomes G. *Nature*, 2023, 624: 570–578
- Zhang L, Liang D, Wang Y, Li D, Zhang J, Wu L, Feng M, Yi F, Xu L, Lei L, Du Q, Tang XJ. *Chem Sci*, 2018, 9: 44–51
- Polak MP, Morgan D. *Nat Commun*, 2024, 15: 1569
- M. Bran A, Cox S, Schilter O, Baldassari C, White AD, Schwaller P. *Nat Mach Intell*, 2024, 6: 525–535
- Zheng Z, Zhang O, Borgs C, Chayes JT, Yaghi OM. *J Am Chem Soc*, 2023, 145: 18048–18062
- Lei G, Docherty R, Cooper SJ. *Digit Discov*, 2024, 3: 1257–1272
- Chen C, Maqsood A, Zhang Z, Wang X, Duan L, Wang H, Chen T, Liu S, Li Q, Luo J, Jacobsson TJ. *Cell Rep Phys Sci*, 2024, 5: 102058
- Jablonska KM, Schwaller P, Ortega-Guerrero A, Smit B. *Nat Mach Intell*, 2024, 6: 161–169
- Zheng Z, Zhang O, Nguyen HL, Rampal N, Alawadhi AH, Rong Z, Head-Gordon T, Borgs C, Chayes JT, Yaghi OM. *ACS Cent Sci*, 2023, 9: 2161–2170
- Liu X, Fan K, Huang X, Ge J, Liu Y, Kang H. *Chem Eng J*, 2024, 490: 151625
- Zheng Z, Rong Z, Rampal N, Borgs C, Chayes JT, Yaghi OM. *Angew Chem Int Ed*, 2023, 62: e202311983
- Zheng Z, Alawadhi AH, Chheda S, Neumann SE, Rampal N, Liu S, Nguyen HL, Lin Y, Rong Z, Siepmann JI, Gagliardi L, Anandkumar A, Borgs C, Chayes JT, Yaghi OM. *J Am Chem Soc*, 2023, 145: 28284–28295

- 13 Kang Y, Kim J. *Nat Commun*, 2024, 15: 4705
- 14 Vatsal S, Dubey H. arXiv: 2407.12994. doi: 2407.12994
- 15 Ku AY, Hool A. *Miner Econ*, 2024, doi: 10.1007/s13563-024-00478-3
- 16 Chen B, Zhang Z, Langrené N, Zhu S. *Patterns*, 2025, 6: 101260
- 17 Leng Y, Zhong Y, Gu Z, Li P, Cui H, Li X, Liu Y, Wan J. *ACS Mater Lett*, 2025, 7: 1807–1816
- 18 Wei J, Wang X, Schuurmans D, Bosma M, Ichter B, Xia F, Chi E, Le Q, Zhou D. arXiv: 2201.11903. doi: 2201.11903
- 19 Liu Y, Yang Z, Zou X, Ma S, Liu D, Avdeev M, Shi S. *Natl Sci Rev*, 2023, 10: nwad125
- 20 Liu Y, Ma S, Yang Z, Zou X, Shi S. *J Chin Ceram Soc*, 2023, 51: 427
- 21 Zheng Z, Rampal N, Inizan TJ, Borgs C, Chayes JT, Yaghi OM. *Nat Rev Mater*, 2025, 10: 369–381
- 22 Li X, Xu W, Zhi C. *Nat Rev Chem*, 2024, 8: 359–375
- 23 Lin D, Li Y. *Adv Mater*, 2022, 34: e2108856
- 24 Liu T, Lei C, Wang H, Li J, Jiang P, He X, Liang X. *Adv Mater*, 2024, 36: 2405473
- 25 Li W, Xu H, Zhang H, Wei F, Zhang T, Wu Y, Huang L, Fu J, Jing C, Cheng J, Liu S. *Energy Environ Sci*, 2023, 16: 4502–4510
- 26 Lv S, Fang T, Ding Z, Wang Y, Jiang H, Wei C, Zhou D, Tang X, Liu X. *ACS Nano*, 2022, 16: 20389–20399
- 27 Zhao J, Chen Y, Zhang M, An Z, Nian B, Wang W, Wu H, Han S, Li Y, Zhang L. *Adv Sci*, 2025, 12: e2410988
- 28 Zheng Z, He Z, Khattab O, Rampal N, Zaharia MA, Borgs C, Chayes JT, Yaghi OM. *Digit Discov*, 2024, 3: 491–501
- 29 Liu S, Wen T, Pattamatta ASLS, Srolovitz DJ. *Mater Today*, 2024, 80: 240–249
- 30 Wang C, Ji X, Liang J, Zhao S, Zhang X, Qu G, Shao W, Li C, Zhao G, Xu X, Li H. *Angew Chem Int Ed*, 2024, 63: e202403187
- 31 Zou Y, Liu T, Du Q, Li Y, Yi H, Zhou X, Li Z, Gao L, Zhang L, Liang X. *Nat Commun*, 2021, 12: 170
- 32 Jiang P, Du Q, Lei C, Xu C, Liu T, He X, Liang X. *Chem Sci*, 2024, 15: 3357–3364
- 33 Li X, Li M, Huang Z, Liang G, Chen Z, Yang Q, Huang Q, Zhi C. *Energy Environ Sci*, 2021, 14: 407–413
- 34 Wang M, Meng Y, Sajid M, Xie Z, Tong P, Ma Z, Zhang K, Shen D, Luo R, Song L, Wu L, Zheng X, Li X, Chen W. *Angew Chem Int Ed*, 2024, 63: e202404784
- 35 Bi S, Wang H, Zhang Y, Yang M, Li Q, Tian J, Niu Z. *Angew Chem Int Ed*, 2023, 62: e202312982
- 36 Hu T, Zhao Y, Yang Y, Lv H, Zhong R, Ding F, Mo F, Hu H, Zhi C, Liang G. *Adv Mater*, 2024, 36: e2312246
- 37 Zong W, Li J, Zhang C, Dai Y, Ouyang Y, Zhang L, Li J, Zhang W, Chen R, Dong H, Gao X, Zhu J, Parkin IP, Shearing PR, Lai F, Amine K, Liu T, He G. *J Am Chem Soc*, 2024, 146: 21377–21388
- 38 Zhang Q, Ma Y, Lu Y, Zhou X, Lin L, Li L, Yan Z, Zhao Q, Zhang K, Chen J. *Angew Chem Int Ed*, 2021, 60: 23357–23364
- 39 Zhou X, Zhao F, Guo Y, Rosenberger B, Yu G. *Sci Adv*, 2019, 5: eaaw5484
- 40 Liu T, Lei C, Wang H, Yang W, He X, Liang X. *Chem Commun*, 2024, 60: 7447–7450
- 41 Ma W, Liu T, Xu C, Lei C, Jiang P, He X, Liang X. *Nat Commun*, 2023, 14: 5508
- 42 Zhang C, Shin W, Zhu L, Chen C, Neuefeind JC, Xu Y, Allec SI, Liu C, Wei Z, Daniyar A, Jiang J-, Fang C, Alex Greaney P, Ji X. *Carbon Energy*, 2020, 3: 339–348
- 43 Xiao T, Yang J-, Zhang B, Wu J, Li J, Mai W, Fan HJ. *Angew Chem Int Ed*, 2024, 63: e202318470
- 44 Wu H, Hao J, Zhang S, Jiang Y, Zhu Y, Liu J, Davey K, Qiao SZ. *J Am Chem Soc*, 2024, 146: 16601–16608
- 45 Liu T, Lei C, Wang H, Xu C, Ma W, He X, Liang X. *Sci Bull*, 2024, 69: 1674–1685